

In the following exercises, all coordinates and components are given with respect to an orthonormal frame $\mathcal{K} = (O, \mathcal{B})$. For the 2-dimensional cases $\mathcal{B} = (\mathbf{i}, \mathbf{j})$ and for the 3-dimensional cases $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$.

1. (1 point) Select all that apply. Which of the following are isometries?
 A. glide-reflections B. all homotheties C. the identity map D. contractions
2. (1 point) Select all that apply. Which of the following are direct isometries?
 A. rotations
 B. compositions of two reflections
 C. reflections composed with translations
 D. glide-reflections

3. (1 point) Select all that apply. Which of the following describe glide-reflections?

A. $\frac{1}{5} \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ B. $\frac{1}{5} \begin{bmatrix} 4 & 3 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ C. $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ D. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

Setup 1: In $\mathbb{E}^2$, consider the point $P(0, 1)$, the line $\ell : x + 2y - 1 = 0$, the orthogonal reflection $\text{Ref}_{\ell}^{\perp}$ in ℓ , and the counterclockwise rotation $\text{Rot}_{O, 45^\circ}$ around the origin O . Moreover, let ϕ be the transformation obtained by applying the rotation after the reflection, i.e.,

$$\phi = \text{Rot}_{O, 45^\circ} \circ \text{Ref}_{\ell}^{\perp}.$$

4. (1 point) Consider Setup 1 above. The point $\text{Ref}_{\ell}^{\perp}(P)$ is
 A. $(-\frac{2}{5}, \frac{1}{5})$ B. $(\frac{-1}{5}, \frac{1}{5})$ C. $(\frac{2}{5}, \frac{-1}{5})$ D. $(\frac{1}{5}, \frac{-1}{5})$
5. (1 point) Consider Setup 1 above. The coordinates of the point $\phi(O)$ are _____.
6. (1 point) Consider Setup 1 above. The inverse transformation ϕ^{-1} of ϕ is $\phi^{-1}(\begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} - & - \\ - & - \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} - \\ - \end{bmatrix}$.

Setup 2: In $\mathbb{E}^2$, consider the points $P(2, 1)$ and $C(-1, 1)$, and the counterclockwise rotation $\text{Rot}_{C, \theta}$ around C by an angle θ where $\cos(\theta) = \frac{3}{5}$. Moreover, let ϕ be the transformation obtained by applying $\text{Rot}_{O, \tilde{\theta}}$ (where O is the origin and $\tilde{\theta} \in \mathbb{R}$) after $\text{Rot}_{C, \theta}$, i.e.,

$$\phi = \text{Rot}_{O, \tilde{\theta}} \circ \text{Rot}_{C, \theta}.$$

7. (1 point) Consider Setup 2 above. Then $\text{Rot}_{C, \theta}(\begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} - & - \\ - & - \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} - \\ - \end{bmatrix}$.
8. (1 point) Consider Setup 2 above. If $\tilde{\theta} = -\theta$, the coordinates of the point $\phi(P)$ are _____.
9. (1 point) Consider Setup 2 above. Select all that apply. If $\tilde{\theta} = -\theta$, the map ϕ is
 A. a translation B. a direct isometry C. not an isometry D. a rotation

Setup 3: In $\mathbb{E}^2$, consider the points $C_1(1,0)$ and $C_2(-1,0)$, and the counterclockwise rotation $\text{Rot}_{C_1,\theta}$ around C_1 by an angle θ where $\cos(\theta) = \frac{3}{5}$. Moreover, let ϕ be the transformation obtained by applying $\text{Rot}_{C_2,\tilde{\theta}}$ (for some $\tilde{\theta} \in \mathbb{R}$) after $\text{Rot}_{C_1,\theta}$, i.e.,

$$\phi = \text{Rot}_{C_2,\tilde{\theta}} \circ \text{Rot}_{C_1,\theta}.$$

10. (1 point) Consider Setup 3 above. If O denotes the origin and $\tilde{\theta} = 90^\circ$, then $\phi(O)$ is _____.
11. (1 point) Consider Setup 3 above. If $\tilde{\theta} = 90^\circ$, the map ϕ is a rotation with center _____.
12. (1 point) Consider Setup 3 above. For which values of $\tilde{\theta}$ is ϕ a translation?
 A. $\tilde{\theta} = -\theta$ B. $\tilde{\theta} = 2\theta$ C. $\tilde{\theta} = \theta$ D. there is no such value

Answers:

1. A, C
2. A, B
3. B, D
4. A
5. $\left(-\frac{\sqrt{2}}{5}, \frac{3\sqrt{2}}{5}\right)$
6. $\phi^{-1}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \frac{\sqrt{2}}{10} \begin{bmatrix} 7 & -1 \\ -1 & -7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \frac{2}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
7. $\text{Rot}_{C,\theta}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \frac{1}{5} \begin{bmatrix} 2 \\ 6 \end{bmatrix}$
8. $\left(\frac{16}{5}, \frac{7}{5}\right)$
9. A, B
10. $\left(\frac{-1}{5}, \frac{7}{5}\right)$
11. $\left(\frac{-1}{3}, \frac{2}{3}\right)$
12. A